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BE (8th Semester)
Examination, April - May, 2017

[New Scheme]

Robotics

Time Allowed : 3 hours *Maximum Marks :* 80
Minimum Pass Marks : 28

Note : (*i*) All questions are compulsory. Part (*a*) of each question is compulsory. Attempt any ***two*** parts from (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

UNIT - I

1. (*a*) Write down laws of robotics. [2]

(*b*) Write short notes on : [7]

(*i*) Robot Anatomy

(*ii*) Design and control issues

(*c*) Explain in detail the Notation used in Robotics and future application of Robotics. [7]

(*d*) Explain different types of Arm configuration with neat sketch. [7]

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UNIT - II

2. (*a*) What is Mapping? [2]

(*b*) Explain D-H notation in detail. [7]

(*c*) Briefly describe the various robot components. Define degree of freedom. How do you calculate the DOF of robot manipulator? [7]

(*d*) What do you understand by global and local coordinate? What do you understand by coordinate frames? [7]

UNIT - III

3. (*a*) What is Forward and Inverse Kinematics? [2]

(*b*) What is composite transformation matrix? Explain the rules to be followed in the formation of the same. [7]

(*c*) Give brief introduction of inverse kinematic model. [7]

(*d*) Explain kinematic model of robotic manipulators with example. [7]

UNIT - IV

4. (*a*) Write down major function of sensors in robot. [2]

(*b*) Explain various kinds of sensor used in robotics. Explain in detail. [7]

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(*c*) What do you understand by the term **Image** acquisition in Robotics? **[7]**

(*d*) Write short notes on : **[7]**

(*i*) Architecture of Robotic vision system

(*ii*) Description of other components of vision system

UNIT-V

5. (*a*) What is main Robots Application? [2]

(*b*) Write short notes on : [7]

(*i*) Robot safety

(*ii*) Non-industrial application

(*c*) Explain principles for robot application and application planning. [7]

(*d*) How can you justify that the robot is better for Industry? Enumerate any four points. [7]

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